Department of Computer Science and Engineering

PROJECT REPORT ON

CURASYNC

– A DOCTOR APPOINTMENT APP

Course name: Mobile Programming
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Submitted to :

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Name	ID	Contribution
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Overview / Introduction

Motivation behind the Project

Healthcare is one of the most essential services for every human being, yet in many developing countries the system of accessing medical services is often inefficient and inconvenient. In Bangladesh, for example, patients frequently struggle to find qualified doctors when needed, especially in specialized areas like cardiology, neurology, or pediatrics. In rural areas, the situation is even more challenging, as most specialist doctors are concentrated in major cities like Dhaka or Chattogram. Patients living in smaller towns or villages are forced to travel long distances, wait for hours, and often end up without a confirmed appointment.

The COVID-19 pandemic further highlighted the importance of digital healthcare. Many people avoided hospitals due to fear of infection, and doctors themselves looked for safer ways to consult patients. This situation proved the need for a **digital appointment booking system** that could serve both patients and doctors while reducing physical contact and saving time.

The motivation behind CuraSync was to create a **mobile-first, user-friendly healthcare platform** that:

- Enables patients to book appointments without leaving home.
- Provides flexibility in consultation types (video, audio, chat, or physical visit).
- Allows doctors to manage schedules and professional details digitally.
- Builds a healthcare ecosystem where both patients and doctors benefit equally.

Challenges Faced During Development

Developing CuraSync was not without challenges. The most significant ones were:

1. **Firebase Connection Setup** – Initially, trying to run the app without updating the Google `google-services.json` file from Firebase. This caused the app to get stuck in the sign-up section with an indefinite wait time, even though the user information was already visible in Firebase Authentication.

2. **Database Management** – The app required a structured database system where patients, doctors, and appointments were all linked. Designing `BookAppointmentDB.java` in a way that allows it to store appointments locally while also preparing for cloud integration was a challenging task.

   Addtionally, retrieving doctor information from Firebase `"Doctors"` and fetching the current user's details from the `"User"` database to populate the "Book an Appointment"

page proved to be quite challenging. The available time slots and days of a doctor were stored as ArrayLists and Map values, which made handling them tricky. Moreover, the same values retrieved from Firebase also had to be stored in SQLite, adding another layer of complexity.

3. **User Interface Consistency** – With over a dozen XML layout files (`activity_book_appoiment.xml`, `activity_home.xml`, `activity_signup.xml`, `activity_login.xml`, `activity_doctor_list_show.xml`, `doctor_item.xml`, `activity_welcome_screen.xml` etc.), maintaining a consistent theme (color scheme, typography, margins) across all screens was essential. This required detailed design guidelines.

4. **Book an Appointment Gridview** – Initially, the available time slots for a doctor were supposed to be retrieved from the database and displayed on the "`Book Appointment`" page in a `GridView`, categorized into morning, afternoon, evening times. However, since the appointment layout was inside ScrollView, the `GridView` did not function properly. It failed to display any data unless set manually, even though the retrieved information was visible in Logcat. Later, this issue was resolved by replacing the `GridView` with a `Linear layout` that dynamically adds `TextViews` to display the available time slots.

5. **Doctor List Page –** The `DoctorListShow.java` page was implemented using an adapter for `doctor_item.xml` along with its own layout file. However, I accidentally assigned a `click listener` to the adapter's view itself, which caused the app to crash repeatedly. Even after removing the click listener, the crashes continued because the adapter's view hierarchy was still being referenced incorrectly, leading to null pointer exceptions or invalid view accesses during runtime.

6. **Scalability Considerations** – From the beginning, the system had to be designed for future features like payments, telemedicine API integration, and push notifications. Writing modular code that allowed for future scalability required careful planning.

Benefit / Usability

The benefits of CuraSync can be considered from three perspectives: **patients, doctors, and the overall healthcare ecosystem**.

Benefits for Patients

- **Convenience**: Patients no longer need to travel and wait in queues to book an appointment. They can log in, browse available doctors, and book instantly.
- **Multiple Consultation Options**: Patients can choose between video consultation, audio consultation, conversational chat, or visiting the doctor in person depending on their need.

- **Transparency**: Each doctor's profile shows specialization, experience, consultation fees, availability, and reviews, helping patients make informed choices.
- **Healthcare Record Keeping**: Appointment history allows patients to maintain a digital healthcare record, which can be useful for follow-up consultations.
- **Accessibility**: Patients in rural areas or those with mobility issues can easily consult specialists through online modes.

Benefits for Doctors

- **Professional Profile Management**: Doctors can register with their BMDC registration number, specialization, and work experience, building credibility.
- **Flexible Scheduling**: They can define availability, consultation fees, and working hours.
- **Expanded Reach**: Through teleconsultation, doctors can serve patients beyond geographical barriers.
- **Organized Workflow**: Appointment requests are digital, reducing the chance of overbooking or mismanagement.

Benefits for Healthcare Ecosystem

- **Efficiency**: Reduces manual work and paperwork.
- **Promotion of Telemedicine**: Encourages online consultations, reducing unnecessary hospital visits.
- **Trust Building**: Verified doctor profiles and transparent consultation fees increase patient trust.
- **Scalability**: The app can be expanded into a larger platform integrating pharmacies, diagnostic centers, and insurance companies.

Implementation Details

Key Modules Implemented

1. **Authentication Module**
    - Login screen `(activity_login.xml)` with email, password, "remember me", and "forgot password" features; these features were implemented as Firebase handles them internally.
    - Signup screen `(activity_signup.xml)` collects patient details like name, email, phone, and password.

- Doctor signup (`activity_doc_reg.xml`) includes professional details like BMDC number, specialization, consultation fees, and license upload.
- Welcome screen (`activity_welcome_screen.xml`) acts as an entry point with options for login, signup, guest access, or Google login.

2. **Home & Search Module**
   - Home screen (activity_home.xml) shows categories like cardiology, neurology, pediatrics, etc.
   - Search bar allows patients to find doctors by name, specialization, or location, which will be a future implementation.
   - Quick links for booking an appointment, viewing appointments, contacting support, and hotline numbers.
3. **Doctor Module**
   - Doctor list (`activity_doc_list.xml`) displays doctors with "Book" and "Call" options.
4. **Appointment Module**
   - Booking screens (`activity_book_appoinment.xml` and `activity_appoinment.xml`) collect patient details, problem description, and allow slot and consultation type selection.
   - Appointment item (`appointment_item.xml`) displays doctor, patient, time, and problem summary.

Exceptional Work Done

- Created custom list adapters for dynamic rendering of doctors and appointments. This includes displaying the doctor list, available appointment time slots, and (planned) view appointment functionality using custom adapters.
- Implemented a modular booking flow that collects patient details and consultation preferences in a single interface.
- Integrated Firebase for secure login and SQLite for local storage of appointment data.
- Maintained UI/UX consistency using a unified color scheme with a beige background, teal highlights, and orange accents.
- Developed a dynamic time-slot system for flexible scheduling, where available times are calculated based on the current date and time.
- Implemented a Guest section that allows users to explore the app without logging in, while prompting registration for any interactive features.

Limitations & Future Implementations

- The most noticeable limitation of the app is the absence of a bottom navigation bar, which would have allowed users to navigate more comfortably between sections.
- The "`View Appointment`" section was coded but not fully implemented, and the "`Book Appointment`" flow currently forces users to select a doctor first. This could be improved to allow more dynamic input based on user preference.
- There is no profile menu in the app, which limits personalization and easy access to user information or settings.
- The app lacks push notifications or reminders for upcoming appointments, which could enhance user engagement, even though FCM was implemented by custom cloud message.
- Offline functionality is limited; without an internet connection, users cannot fetch doctor data or appointments in real time.

Business Model

To sustain the platform financially, CuraSync adopts a **hybrid business model** combining commission, subscription, and partnerships:

1. **Commission Model**: A small commission is charged for each paid consultation booked through the app.
2. **Doctor Subscription Plans**: Premium subscriptions for doctors allow them to appear higher in search results, receive detailed analytics, and access advanced features.
3. **In-App Advertising**: Healthcare brands, diagnostic centers, and pharmacies can advertise relevant products and services inside the app.
4. **Value-Added Services**: Integration with pharmacy delivery, diagnostic test booking, and health insurance packages.
5. **Enterprise Licensing**: Hospitals and clinics can license CuraSync as a white-label solution for internal use.
6. **Freemium Model for Patients**: Basic features remain free, while advanced features like unlimited teleconsultations or premium doctors may be offered under subscription.

Conclusion

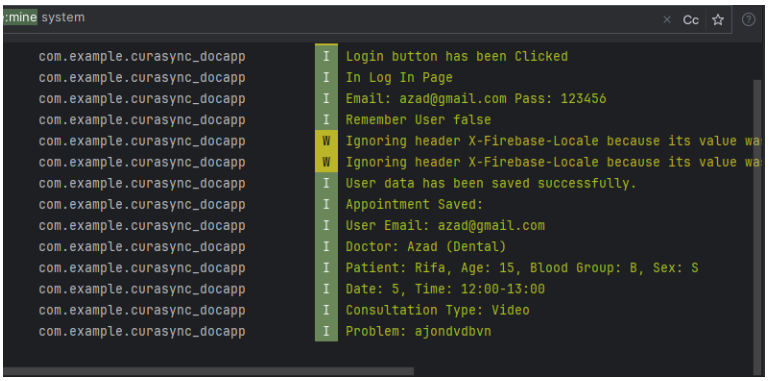
CuraSync is more than just a simple doctor booking app; it is a **digital healthcare platform** designed to improve patient accessibility, doctor convenience, and overall healthcare efficiency. By combining patient and doctor modules with secure authentication, structured database design,

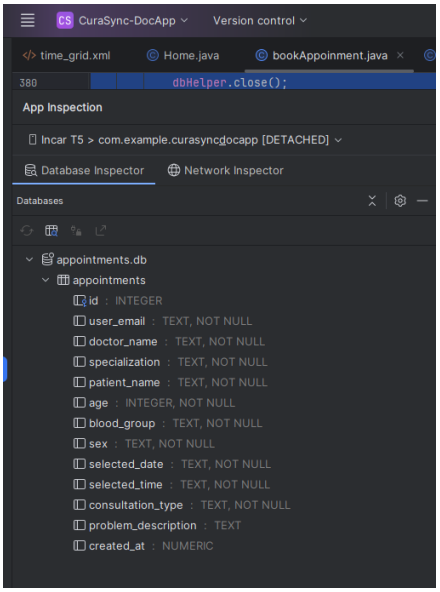
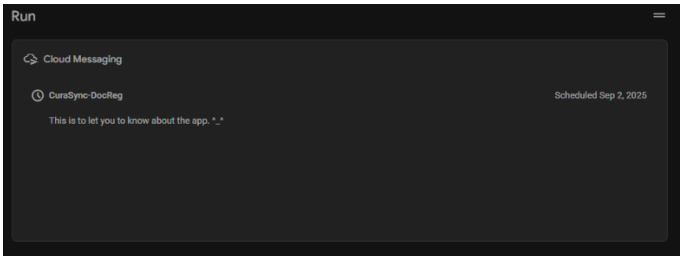
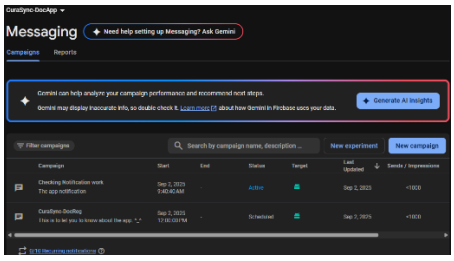
and a user-friendly interface, the app successfully addresses challenges in the existing healthcare system.

The app's modular architecture ensures scalability, allowing future integration of advanced features like payments, video APIs, and AI-driven recommendations. CuraSync demonstrates how technology can directly improve healthcare outcomes in Bangladesh and has the potential to expand internationally.

Screenshots

- Although SQLite implementation cannot be observed in the app right now, for not have "`View Appointment`" page, the screenshots of the app inspection indicate the intended use of a local database for the Appointment class and a `logcat` print to show the data stored in local database.





- For notifications, a custom campaign using Firebase was implemented to show alerts even when the app was not in use. However, users could not see these notifications because the `google-services.json` file was not updated, to avoid the risk of losing existing app data.